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Key Points:

- We reconstruct the Viking 2 seismic workflow and demonstrate how its unique sampling strategy altered recorded seismic signals
- The Viking events potentially located within the northern-plains region later inferred by InSight to be the source of seasonal marsquakes
- Alternative interpretations for the sol 80 event, including a distant tectonic marsquake or a meteorite impact, cannot be excluded

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

J. Li and Y. Wang,
JLI@pku.edu.cn;
ybwang@pku.edu.cn

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Author Contributions:

Conceptualization: Jiaqi Li
Data curation: Wanbo Xiao, Andrew Lazarewicz
Formal analysis: Wanbo Xiao, Jiaqi Li
Funding acquisition: Wanbo Xiao, Jiaqi Li
Investigation: Wanbo Xiao, Jiaqi Li
Methodology: Wanbo Xiao, Jiaqi Li, Andrew Lazarewicz
Project administration: Jiaqi Li
Software: Wanbo Xiao, Andrew Lazarewicz

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Fifty Years Later: Revisiting Viking 2 Seismic Signals Through InSight

Wanbo Xiao[1,2], Jiaqi Li[1], Yanbin Wang[2], and Andrew Lazarewicz[3,4]

[1]SKLab-DeepMinE, MOEKLab-OBCE, School of Earth and Space Sciences, Peking University, Beijing, China, [2]State Key Laboratory of Earthquake Dynamics and Forecasting, Peking University, Beijing, China, [3]The Viking Seismology Team, Pasadena, CA, USA, [4]Retired

Abstract For nearly 50 years, the origin of the candidate seismic signals recorded by Viking 2 remained unresolved because of limited instrumental sensitivity, onboard observation, data compression, and the absence of Martian seismic context. Here we reconstruct the Viking seismic workflow and reinterpret the two Viking candidate events, recorded on sol 53 and sol 80, using the seismic context established by InSight. We show that the Viking seismometer was optimized for high-spectral-frequency seismic signals, consequently reducing its sensitivity to low-spectral-frequency events. If the original 10-s S-P travel-time difference is correct, the inferred location of the sol 80 event overlaps the broad northern-plains source region suggested for the seasonally recurrent high-spectral-frequency and 2.4-Hz marsquake populations observed by InSight. Alternatively, if the first-arrival phase represents a crustal conversion phase rather than a direct P wave, the sol 80 event corresponds to a distant tectonic (low-spectral-frequency or broadband) marsquake with an estimated moment magnitude of M_W 4.6–4.8 or M_W 4.9–5.1. A meteorite impact origin cannot be excluded either. The two Viking seismic signals decay more rapidly than typical InSight marsquakes, suggesting heterogeneity in the Martian attenuation structure. Together, these findings integrate Viking observations into the modern framework of Martian seismicity, illustrating how a legacy planetary data set can acquire renewed scientific implications when re-examined in light of subsequent mission data.

Plain Language Summary For nearly 50 years, scientists debated whether the candidate seismic signals recorded by Viking 2 reflected true Martian tectonics, considering limitations of the Viking instrument. Decades later, the substantial marsquakes detected by the InSight mission revealed a clearer seismic context of the planet. Here we reconstruct the original Viking seismic workflow and re-examine its candidate events, recorded on sol 53 and sol 80, using modern knowledge from InSight. The sol 80 event possibly located in the same northern-plains source region inferred for InSight seasonal marsquakes, although a distant tectonic marsquake or a meteorite impact remain as alternative explanations. The faster decay of the sol 80 event than typical InSight marsquakes could imply lateral variations of structural energy absorption in the Martian subsurface. More broadly, this study shows how a historical planetary data set can gain new scientific implications when revisited using later mission data.

1. Introduction

Fifty years after the Viking 1 and 2 landers touched down on Mars, their missions endure as a landmark in planetary exploration. Long recognized for landmark missions to search for extraterrestrial life (Klein, 1977; Martin, 1969; Panagakos & Parker, 1976), NASA's Viking program also undertook a groundbreaking yet ultimately enigmatic endeavor: the first seismic survey of the Red Planet (Anderson et al., 1972, 1976). The integration of seismometers underscored a profound ambition to unravel the planet's internal structure and tectonic activity—key proxies for its geological evolution and habitability potential (Giardini et al., 2020; Lognonné et al., 2020, 2023).

The ambition to listen to another world's seismic vibrations drew on seismology's proven track record. For over a century, seismic studies on Earth have been the primary tool for deciphering our planet's subsurface structure and dynamics. This approach was successfully adapted for lunar seismology by the Apollo missions in 1969–1977 (Garcia et al., 2011, 2019; Latham et al., 1969; Nakamura et al., 1982; Nunn et al., 2020). Their experiments mapped the Moon's internal layers and seismicity for the first time, establishing the fundamental framework for planetary seismology. Viking sought to extend this legacy to one more world: Mars.

Supervision: Jiaqi Li, Yanbin Wang
Validation: Jiaqi Li, Andrew Lazarewicz
Visualization: Wanbo Xiao, Jiaqi Li
Writing – original draft: Wanbo Xiao, Jiaqi Li
Writing – review & editing: Wanbo Xiao, Jiaqi Li, Andrew Lazarewicz

Yet probing the seismic activity of Mars proved challenging during the Viking era. The Viking 1 seismometer failed to unlock its sensor, and the Viking 2 seismometer (landed in Utopia Planitia), mounted directly on the lander deck, was largely compromised by wind-induced noise (Anderson et al., 1972, 1976; Moore et al., 1977). Consequently, its record of 560 Martian solar days (sols) was deemed seismically barren, yielding only two candidate seismic signals (Anderson et al., 1976). For half a century, the Viking 2 seismic records indicated less seismicity than Earth (Figure 1), and in-depth seismological investigations of Martian tectonics and interiors were restricted by its low sensitivity and strong environmental contamination (Goins & Lazarewicz, 1979).

The recent InSight mission to Mars (2018–2022) ultimately resolved these decades-long ambiguities (Banerdt et al., 2020; Giardini et al., 2020; Knapmeyer et al., 2023). Learning from Viking's lessons, InSight deployed its seismometer directly on the ground and employed a sophisticated wind and thermal shield via its robotic arm (Lognonné et al., 2019). This strategy uncovered a seismically active planet, identifying over 1,300 marsquakes and impacts (Ceylan et al., 2022; InSight Marsquake Service, 2023), most of which detected during the very quiet nighttime hours (Lognonné et al., 2020, 2023). InSight's success raises a critical new question: if the two seismic candidate signals recorded by Viking 2 were indeed marsquakes, how similar are they to the marsquakes later detected by InSight? Answering this question would not only test the seismic origin of the Viking signals, but also probe whether Martian tectonic activity persisted over nearly half a century.

Prior to InSight, interpretations of Viking signals relied on terrestrial analogs (Anderson et al., 1976, 1977). Today, with InSight's seismic waveform data (InSight Mars SEIS Data Service, 2019a, 2019b) and seismic catalog (InSight Marsquake Service, 2023) as definitive references, we know that marsquakes exhibit distinct spectral characteristics (Ceylan et al., 2021, 2022; Clinton et al., 2021; Dahmen et al., 2021; Driel et al., 2021), alongside a basic 1D model of Martian crustal structure (Kim et al., 2022; Knapmeyer-Endrun et al., 2021; Li et al., 2022) and internal scattering and intrinsic attenuation properties (Lognonné et al., 2020; Menina et al., 2021, 2023). Furthermore, the Viking instrument's unique data processing workflow returned compressed and downsampled data rather than raw waveforms (Anderson et al., 1976, 1977; Lazarewicz, 2023; Lazarewicz et al., 1975). Its underlying algorithms can be rewritten with contemporary coding to make them available to the broader seismic community. To date, no study has comprehensively analyzed the Viking seismic data set nor detailed its operational sampling workflow.

On this 50th anniversary of the Viking landings, and at the dawn of a new era of lunar seismic exploration such as the FSS lunar seismometers of NASA's CP-12 mission (Panning et al., 2024), the Lunar Environment Monitoring Station suite (DellaGiustina et al., 2025) and South Pole Seismic Station (Knapmeyer-Endrun et al., 2026) of upcoming Artemis lunar landing missions, and the lunar seismograph of China's Chang'e-7 (C. Wang et al., 2024), we comprehensively synthesize, reconstruct, and archive the Viking seismic workflow and data set. Leveraging this recovered foundation, we bridge observations between the Viking and InSight eras and re-evaluate the origins and mechanisms of the candidate seismic signals on Viking sols 53 and 80.

This effort comes at a key inflection point: five decades later, we are lucky that key original Viking participants are still with us to share irreplaceable firsthand context, making now the last viable window. This reappraisal secures the scientific legacy of a historic mission. By revisiting this five-decade-old data set, our work honors Viking's pioneering legacy and delivers a second data point for Martian seismology, extending InSight's findings across space and, uniquely, back in time.

2. Data and Method

2.1. Viking 2 Data Sets

The Viking 2 seismometer comprises three components: the nearly vertical X component and two horizontal Y, Z components. The seismometer operated in three modes, down-sampling the raw seismic data from an original sampling rate of 121.12 Hz (time increment 8.25e−3 second) to three different rates before returning (Anderson et al., 1976, 1977): High-data-rate mode (20.2 Hz), Event mode (1.01 Hz), and Normal mode (0.067 Hz). The Normal mode was designed to characterize the background noise, while the Event mode could be triggered either by Earth command or automatically by a rapid amplitude increase (Anderson et al., 1976, 1977). During the solar conjunction (sols 64–110), Event-mode records consistently appear between local time 3:00 and 4:00, indicating a manually scheduled Event-mode operation during this interval (Figure 1b), while the small number of Event-mode segments outside this window could have been auto-triggered detections.

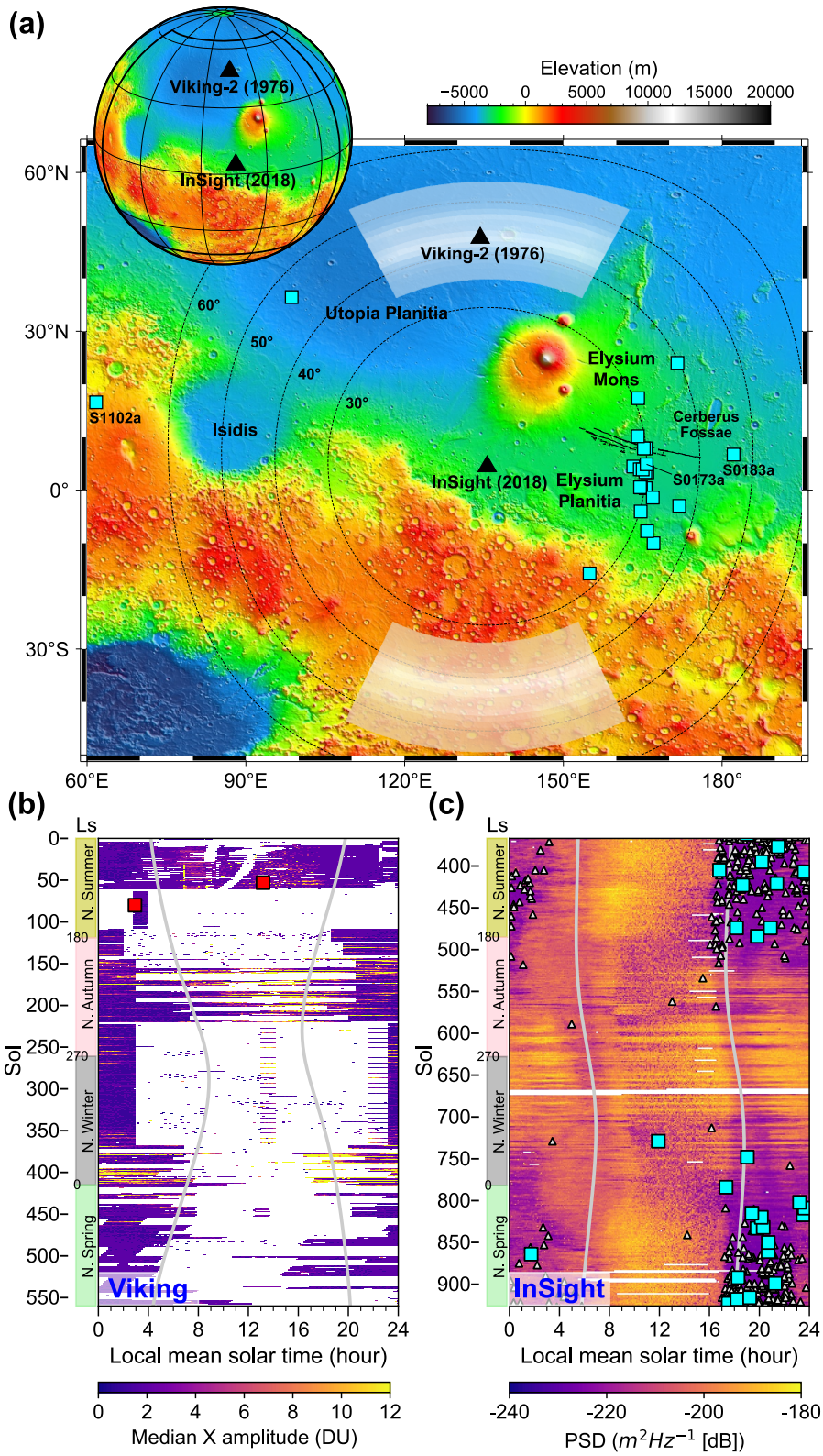

(a)
Viking-2 (1976)
InSight (2018)
Elevation (m)
−5000 0 5000 10000 15000 20000
60°N
30°N
0°
30°S
60°E
90°E
120°E
150°E
180°
Viking-2 (1976)
Utopia Planitia
60°
50°
40°
30°
Elysium Mons
Cerberus Fossae
S1102a
Isidis
InSight (2018)
Elysium Planitia
S0173a
S0183a
(b)
Ls
Sol
N. Summer
N. Autumn
N. Winter
N. Spring
Viking
Local mean solar time (hour)
Median X amplitude (DU)
(c)
InSight
PSD (m^2Hz^{-1} [dB])

Figure 1.

The returned Viking 2 seismic data underwent multiple stages of reformatting (Lorenz et al., 2017), transitioning from cassette binary tapes to structured ASCII digital files. Comparison between independent data sets suggests that the University of Texas Institute for Geophysics data set (hereafter UTIG; see Open Research section) is the most reliable (Text S1 and Figures S1–S3 in Supporting Information S1). We thus used the UTIG data set, whose three-mode recordings almost cover the entire observation period in sols 0–560 (Figure 1b and Figure S4 in Supporting Information S1). Apart from seismic data sets, meteorological measurements of wind speed, pressure, and air temperature during the Viking 2 mission are available on PDS (Figure S5 in Supporting Information S1), though at a much lower sampling rate.

Due to the limited Mars-Earth communication bandwidth, Viking 2 compressed the raw displacement records to amplitude and zero-crossing (i.e., the number of times the signal crosses the *x*-axis, a metric that encodes signal frequency) values prior to transmission. Statistical analyses of the amplitude values show that one standard deviation above the median ambient noise level of the whole mission is around 10 digital units (DUs; Text S2 and Figure S9 in Supporting Information S1).

For the two Viking candidate events on sol 53 and sol 80, we obtain continuous seismic records from the UTIG data set by sequentially merging data blocks with adjacent start times. The amplitude values are moved forward by one sample to compensate for the time delay caused by the Viking 2 digital filter (Texts S3–S4 and Figure S11 in Supporting Information S1).

2.2. Viking 2 Workflow: Simulation and Waveform Reconstruction

The key procedures of Viking 2 seismometer simulation are to convolve displacement waveforms with the seismometer response and then extract the amplitude and zero-crossing values from the convolved waveforms after smoothing (Text S5 and Figures S12–S14 in Supporting Information S1).

Following the original Fortran implementation of Viking 2 simulation, we developed a new Python-based codebase for the Viking 2 seismometer workflow. For the convolution with response, we recovered the instrument response of Viking 2 seismometer using the filtering coefficients in the Fortran documentation codes (Lazarewicz, 2023) and optimized the convolution operation in the frequency domain (Text S3 in Supporting Information S1). The extraction of Event-mode amplitude and zero-crossing values is conducted for every 120-point segment (total length of 0.99 s).

The waveform reconstruction is thus almost the converse process of the seismometer simulation. Every amplitude and zero-crossing values are used to reconstruct a data series of 120 points, under the assumption of a sinusoidal signal. This strategy recovers the signal with high spectral frequency content (Figure S15 in Supporting Information S1). In the reconstruction process, we make linear interpolation for the amplitude value and ensure that the total phase change period is controlled by the zero-crossing value. More technical details are described in Text S6 in Supporting Information S1.

We utilize a sinusoidal signal as an example to show how this workflow functions (Figure S15 in Supporting Information S1). The input signal consists of a low-frequency segment at 0.6 Hz and a high-frequency segment at 3.0 Hz. The output amplitudes exhibit considerable difference in magnitude, stemming from the frequency-dependent sensitivity of Viking 2 response. Output zero-crossing values cycle between integers, with the

Figure 1. Overview of Martian seismic missions (Viking 2 and InSight). (a) Geological context, mission locations, and InSight marsquake distribution. Mars' topographic map is plotted using data from Mars Global Surveyor Laser Altimeter mission (Smith et al., 2003). Black segments represent Cerberus Fossae fault system (Perrin et al., 2022). Cyan cubes present the reliable locations of tectonic marsquake (i.e., low-spectral-frequency and broadband events, excluding meteorite impacts) from InSight Marsquake Service (2023) catalog (Text S9 in Supporting Information S1). Location distributions of InSight high-spectral-frequency marsquakes are shown in white shadow zones (Dahmen et al., 2026). (b) Median *X*-component (nearly vertical) amplitude records of Viking 2 observations, for the Event-mode data blocks of the University of Texas Institute for Geophysics data set (https://darts.isas.jaxa.jp/en/datasets/darts:viking-utig-data). Viking 2's marsquake candidates are shown in red cubes. (c) Average power spectral densities of InSight observations for vertical displacements over the frequency range 0.05–4.0 Hz. Cyan cubes represent InSight's tectonic marsquakes within the presented sol range. Triangles represent InSight high-spectral-frequency swarm marsquakes (Dahmen et al., 2026). White areas are due to data gaps. Ls denotes solar longitude at the Viking 2 and InSight landing sites. InSight data are aligned to Viking 2 data in terms of solar longitude (Ls), covering approximately one Martian year. Background noise is closely related to wind conditions. For Viking 2, wind is strongest between local mean solar time (LMST) 8 and 22 hr during autumn and winter (sols 120–420 here; Figure S4a in Supporting Information S1). For InSight, strong wind is between LMST 6 and 18 hr in spring and summer (sols 370–480 and 780–930).Thick gray curves mark local sunrise and sunset times. A sol, or one Martian day, has a total duration of 24 hr 39 min 35.24 s in Earth time, divided to 24 LMST hours.

average aligned to the input central frequency. The reconstructed waveform and spectrogram can basically reproduce the characteristics of the high-frequency segment, while the low-frequency segment is poorly recovered. More simulation examples are presented in Figures S16–S23 in Supporting Information S1, including InSight's seasonal high-spectral-frequency (HF) marsquake S0260a, tectonic low-spectral-frequency (LF) and broadband (BB) marsquakes (S0173a, S1102a, S1133c) and site-effect-amplified marsquake S1222a, and very-high-spectral-frequency (VF) marsquakes and impacts (S0976b, S0986c, and S1094b). Consistent with the sinusoidal example, the low-spectral-frequency event S0173a exhibits a significantly degraded signal-to-noise ratio, whereas other events with high-spectral-frequency energy remain identifiable.

These Viking simulations convert InSight's displacement waveforms into Viking DU records, effectively reproducing the signals that would have been recorded had a Viking 2 seismometer been deployed at the InSight landing site (Text S5 in Supporting Information S1). The simulations account for the Viking instrument response, while instrumental self-noise is evaluated separately (Text S8 in Supporting Information S1). Our estimates show that the Viking seismometer's self-noise floor is comparable to that of the InSight's Short Period sensor at the natural frequency (4 Hz) and remains well below 1 DU (Text S8 in Supporting Information S1), although it is approximately an order of magnitude higher than that of the InSight's Very Broad Band sensor in the 0.01–1 Hz band (Lognonné et al., 2019; Mimoun et al., 2017).

The reconstructed waveform, due to the incomplete recovery of the original spectral shape, is unsuitable for the application of the spectral fitting method for magnitude estimation as for InSight events (Text S9 in Supporting Information S1). We have therefore adopted a magnitude estimation method based on waveform amplitude scaling. We first convert the InSight waveforms into simulated Viking 2 Event-mode amplitudes. We then compare the sol 80 amplitude with those of InSight marsquakes of known magnitude and epicentral distance. The resulting amplitude ratio, together with a distance correction based on the magnitude scaling relation derived for InSight marsquakes (Böse et al., 2021), is used to estimate the magnitude of the sol 80 event at a given epicentral distance.

2.3. Performance of Viking 2 Seismometer: A 1970s Design

The effective frequency range of a planetary seismometer was largely restricted by the engineering limitations on mass and power (Lognonné & Mosser, 1993; Lognonné et al., 1998). Legacy seismometers, such as the Viking instrument, were characterized by narrowband peak sensitivity at high spectral frequencies (i.e., Fourier frequencies of the recorded signal, rather than event occurrence rates), in contrast to modern broadband seismometers, which maintain high gain across wide frequency ranges (Lognonné et al., 2019). For example, the Viking 2 seismometer was designed to have a peak sensitivity at 3 Hz, with amplification declining sharply at lower spectral frequencies (Figure 2).

A further critical restriction arose from the Mars-Earth communication bandwidth, which limited the returned seismic data stream to ∼2 samples per second. Per the Nyquist-Shannon theorem (Nyquist, 1928), this rate would aliase any signal content above 1.0 Hz, making the system inherently unable to capture unambiguous waveforms of the high-spectral-frequency signals. To circumvent this limitation, a unique “Viking sampling technique” was developed (Anderson et al., 1976, 1977; Lazarewicz, 2023): the system first sampled data at 121.12 Hz, then resampled to ∼1 Hz to store smoothed amplitudes, alongside derived zero-crossing values to preserve high-spectral-frequency content. Because the zero-crossing parameter records the number of times the signal crosses the x-axis within each sampling interval, signals below ∼1 Hz cannot be adequately represented. Thus, the Viking sampling scheme preferentially preserved high-spectral-frequency content while largely losing low-spectral-frequency seismic information.

This coherent design package, optimized for high-frequency energy, yielded the appearance of an apparently limited marsquake numbers: only two candidate events were identified over its 560-sol operational lifetime. The InSight mission (2018–2022) showed a different view, detecting over 1,300 seismic events in ∼1400 sols (Ceylan et al., 2022; InSight Marsquake Service, 2023). Most of the top 100 largest InSight marsquakes were classified as tectonic marsquakes (low-spectral-frequency and broadband events in the official InSight Marsquake Service catalog). These tectonic marsquakes typically occurred at regional epicentral distances (within ∼35°) and were dominated by low spectral frequencies (typically 0.1–1 Hz), bearing closer spectral resemblance to terrestrial volcanic or slow earthquakes than to typical earthquakes. The operational parameters of the Viking instrument rendered it effectively blind to the low-spectral-frequency energy now recognized as characteristic of tectonic marsquakes.

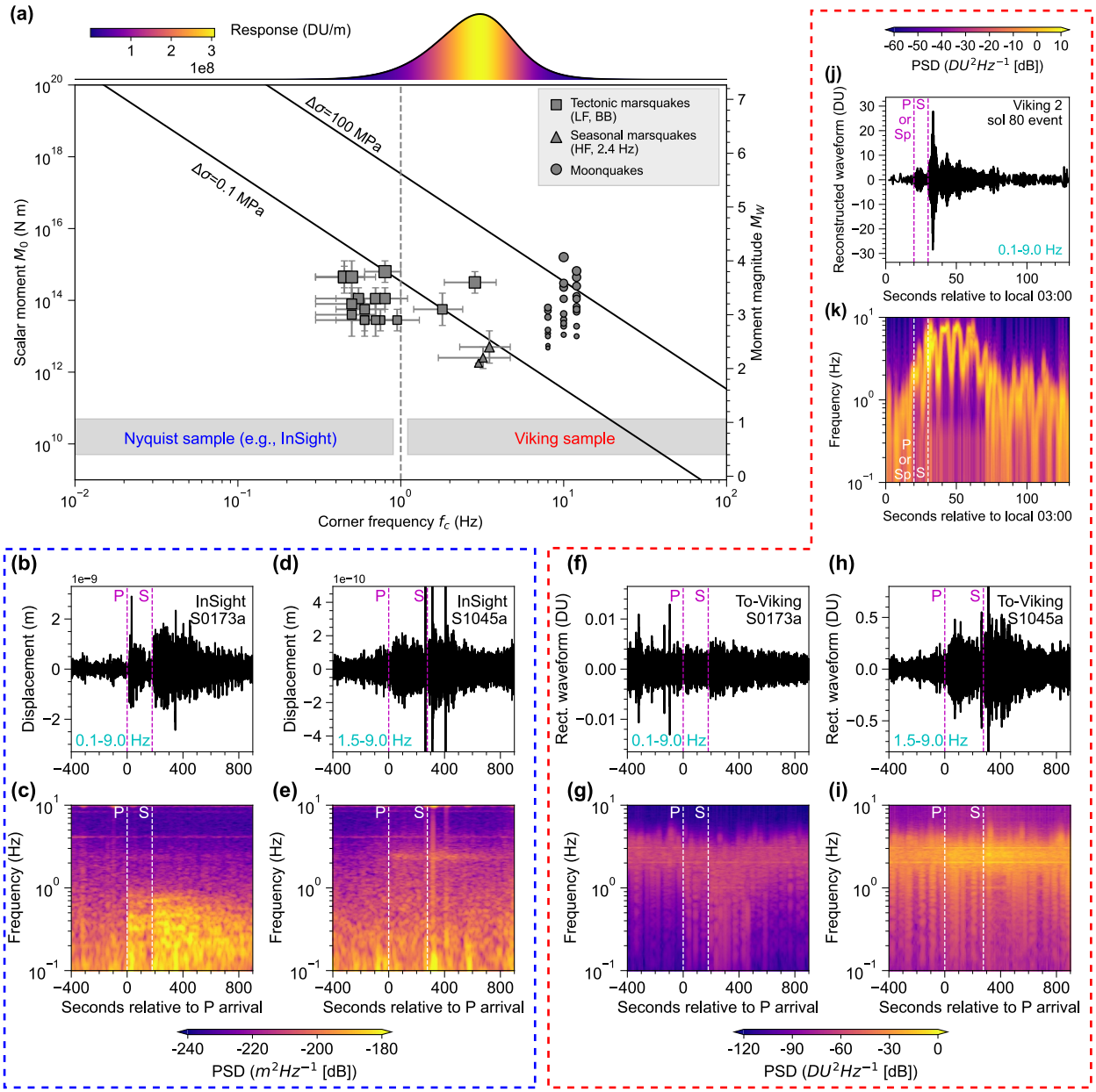


Figure 2. Performance of Viking 2 seismometer sampling workflow. (a) Source parameters of InSight tectonic marsquakes (low-spectral-frequency and broadband events), InSight seasonal marsquakes (high-spectral-frequency and 2.4 Hz events), and Apollo shallow moonquakes (Oberst, 1987). Source parameters for the InSight marsquakes are from corner-frequency analysis of Stähler et al. (2022). The colored curve at the top shows the frequency-dependent response of the Viking 2 seismometer. Nyquist sampling (e.g., InSight) denotes if Viking had taken conventional continuous ∼2 Hz waveform sampling, whereas the actual Viking sample scheme returned compressed ∼1 Hz amplitude and zero-crossing measurements. Most tectonic marsquakes fall outside the frequency ranges effectively captured by both the Viking 2 seismometer response and its sampling scheme, owing to their predominant low-spectral-frequency energy. Dashed vertical line marks 1 Hz. Solid lines indicate constant stress drops of 0.1 and 100 MPa. (b–e) Original InSight displacement waveforms and corresponding spectrograms for the tectonic marsquake S0173a (b, c) and seasonal marsquake S1045a (d, e), rotated to obtain the *X* component (nearly vertical). Dashed lines mark the *P*- and *S*-wave arrivals. (f–i) Corresponding signals after conversion through the Viking 2 instrument response and sampling workflow (Text S5 in Supporting Information S1), shown as reconstructed Event-mode waveforms and spectrograms for S0173a (f, g) and S1045a (h, i), converted to the epicentral distances relative to Viking 2 (2° for S1045a). The detection threshold of the Viking 2 seismometer is 1 digital unit. (j, k) Reconstructed Event-mode waveform and spectrogram of the Viking 2 sol 80 event. Magenta dashed lines indicate the alternative interpretations of the first arrival as either P or Sp and the subsequent S arrival. Time is relative to local 03:00 (10800th Earth second) on Viking 2 sol 80. A sol comprises 24 hr 39 min 35.24 s.

To quantitatively illustrate this performance limitation, we have simulated the digital end-to-end response of the Viking system (neglecting self-noise of the Viking sensors), using the well-characterized, LF InSight event S0173a (M_W 3.7, epicentral distance ∼30°) as the input source (Figure 2). Our modeling reveals a dual failure mode: (a) The sampling algorithm's zero-crossing logic would systematically strip out the event's defining low-spectral-frequency energy; (b) the seismometer's intrinsic peak sensitivity at 3 Hz is spectrally misaligned with S0173a's energy, which is concentrated below 1 Hz. Given that Viking's peak sensitivity was already over three orders of magnitude lower than that of InSight's Very Broad Band seismometer (Lognonné et al., 2019), signals in S0173a's low-spectral-frequency band would be reduced far below Viking's detection threshold (DU at least ≥1). Consequently, tectonic LF events analogous to S0173a were unidentifiable by the Viking system. In contrast, seasonal marsquake (S1045a) and other events with obvious spectral energy up to 2–3 Hz (e.g., VF and impacts) remain identifiable in both amplitude envelopes and reconstructed waveforms (Figure 2 and Figures S16–S23 in Supporting Information S1), because their spectral energy concentrated near 2–3 Hz is close to the Viking seismometer's peak response frequency.

3. Unveiling the Viking 2 Sol 80 Event

3.1. Candidate Seismic Events

The Viking 2 seismometer recorded two anomalous signals on sols 53 and 80, initially cataloged as potential marsquakes. These two events exhibit waveform characteristics that are notably similar to earthquakes recorded in California and marsquakes detected by the InSight mission (Figure 3), yet distinctly different from the identified Viking wind noise (Figures S2 and S3 in Supporting Information S1; Anderson et al., 1977; Lazarewicz, 2023). According to the lander activity catalog (Lorenz et al., 2017), there are no lander operations during these two events.

The sol 80 event emerges as the most credible candidate for a seismic signal (Figure S24 in Supporting Information S1), primarily because its occurrence during the local Martian early morning (local time ∼03:00) aligns with periods of minimal wind (Figure S25 in Supporting Information S1). Although no simultaneous wind record exists for the sol 80 event, the nearest available data, recorded 20 min earlier, show a low wind speed of around 2.6 m/s (Lorenz et al., 2017). Additionally, its peak amplitude (∼30 digital units) substantially exceeds the mission's nighttime ambient noise floor (∼10 digital units; Text S2 in Supporting Information S1). In contrast, the sol 53 event occurred during the noisy afternoon (local time ∼13:30), associated with a higher wind speed of 4.9 m/s, and its amplitude (20–25 digital units) is closer to the surrounding background noise (Figure S26 in Supporting Information S1). Given its higher signal-to-noise ratio and the calmer wind conditions near the event occurrence, we regard the sol 80 event as the clearer representative of a seismic event.

3.2. Seasonal Marsquakes (InSight HF and 2.4-Hz Events)

In 1977, the Viking Team (Anderson et al., 1977) first estimated the epicentral distance of the sol 80 event using its measured S-P travel time (∼10 s; Figure S27 in Supporting Information S1), yielding a value of 110 km (equivalent to ∼2° on Mars). If correct, this interpretation would place the event within the northern plains of Mars and imply a regional marsquake occurring close to the lander. The S-P travel time difference of the sol 53 event (∼33 s; Figure S25 in Supporting Information S1) thus corresponds to a slighter farther distance (∼5°). During the Viking era, however, the lack of a Martian seismic context prevented further interpretation.

The success of InSight fundamentally updates such context. Among the ∼1,300 marsquakes detected by InSight, fewer than 100 correspond to classical tectonic marsquakes with impulsive body-wave arrivals and well-constrained locations, most of which cluster near Cerberus Fossae (InSight Marsquake Service, 2023), a young graben system thought to be among the most tectonically active regions on present-day Mars. In contrast, the dominant seismic population on Mars consists of the high-spectral-frequency (HF) and 2.4-Hz event families, comprising nearly 1,000 events with emergent arrivals, poorly constrained back azimuths, and strong seasonal modulation in occurrence (Knapmeyer et al., 2021). This apparent abundance in the InSight catalog (∼1,300 events), however, may be partly enhanced by the strong 2.4-Hz site amplification beneath the InSight station (Hobiger et al., 2021), an effect that may not be present at the Viking 2 landing site. Although the physical origin of InSight's seasonal events remains debated, their estimated epicentral distances cluster near ∼44°, and their inferred back azimuths preferentially align near the north–south direction, pointing toward a broad source region in the northern plains that encompasses the Viking 2 landing site (Figure 1; Dahmen et al., 2026).

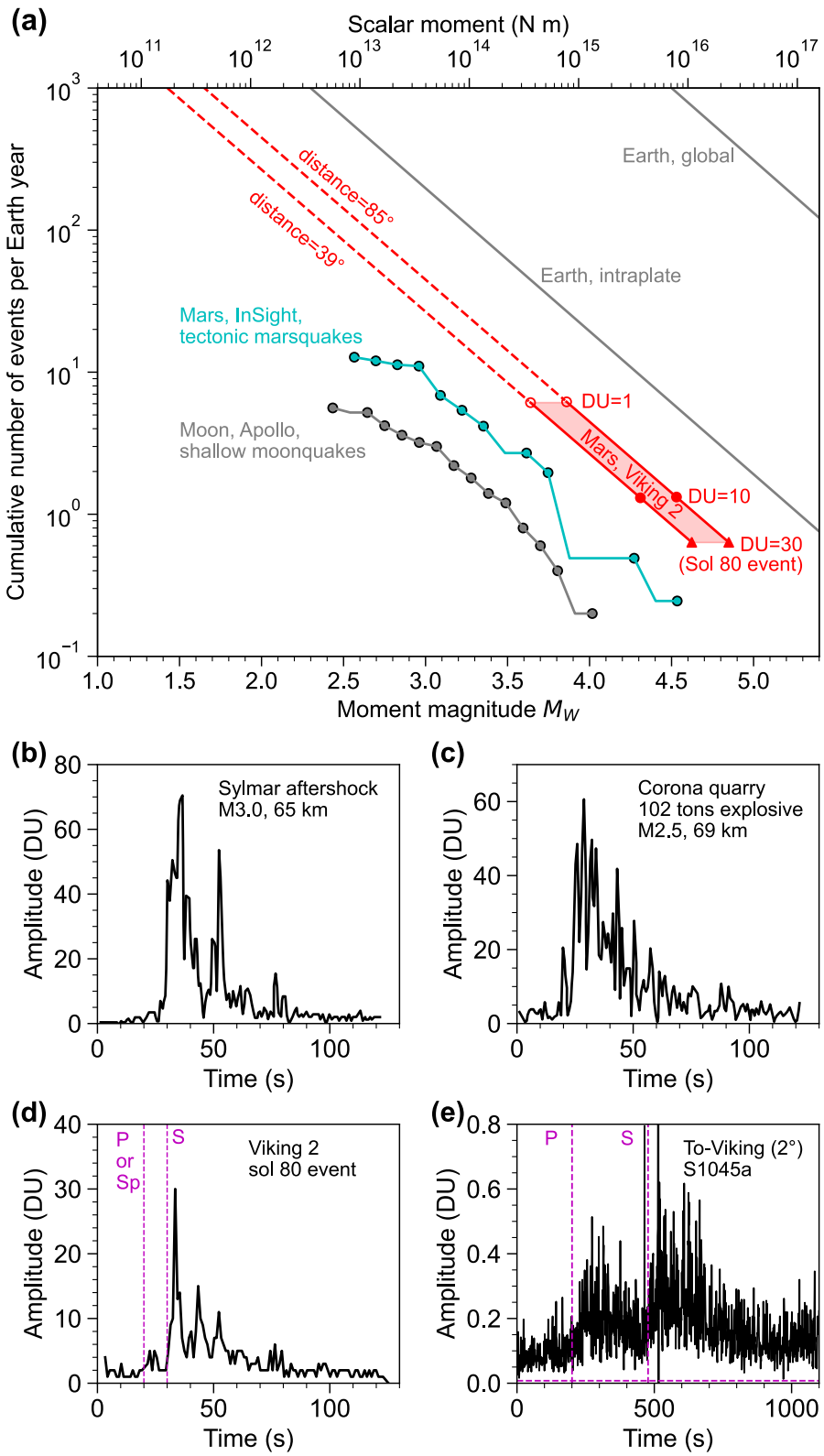

(a)
Scalar moment (N m)
Cumulative number of events per Earth year
Moment magnitude M_W
Earth, global
Earth, intraplate
distance=85°
distance=39°
Mars, InSight,
tectonic marsquakes
Moon, Apollo,
shallow moonquakes
DU=1
Mars, Viking 2
DU=10
DU=30
(Sol 80 event)
(b)
Sylmar aftershock
M3.0, 65 km
Amplitude (DU)
Time (s)
(c)
Corona quarry
102 tons explosive
M2.5, 69 km
(d)
P
or
Sp
S
Viking 2
sol 80 event
(e)
P
S
To-Viking (2°)
S1045a

Figure 3.

Seasonal marsquakes constitute the dominant population in the InSight catalog (~1,000 out of ~1,300 events). Most of these events would fall below Viking's 1-DU detection threshold at their typical epicentral distance of ~44° as recorded at the InSight station (Figure 4). However, their inferred source region lies near the Viking 2 landing site, implying substantially shorter source-receiver distances for Viking 2. Had analogous events occurred within ~2° (~110 km) of the Viking 2 landing site, their amplitudes would have been substantially larger, with a considerable number exceeding the Viking detection threshold (Figure 4). The combination of their abundance, inferred proximity to Viking 2, and high-spectral-frequency content therefore makes seasonal marsquake swarm events among the most plausible marsquake populations detectable by Viking 2.

Moreover, the similarity in seasonal occurrence and epicentral location between the Viking sol 80 event and the InSight HF swarm events strengthens the possibility of a common origin. The Viking 2 sol 80 event was recorded during the local summer, when near-surface temperatures are highest, consistent with the concentration of InSight HF swarm events in the warmer local spring and summer seasons (Figure 1). More strikingly, despite being recorded nearly 50 years earlier by a substantially less sensitive instrument and processed through an entirely different workflow, the independently estimated location of the sol 80 event lies within the same regions inferred for the InSight's seasonal marsquakes.

This independent spatial correspondence provides further support for the Martian northern plains as the source region of the InSight seasonal marsquakes, whose locations remain difficult to constrain because of their emergent seismic phases and ambiguous polarization. It also suggests that present-day seismic activity extends beyond Cerberus Fossae into a separate region. Together with their similar seasonal occurrence, the Viking observation points to the possibility that this seasonally active region has persisted for nearly half a century, potentially linked to climate- or volatile-controlled processes (Knapmeyer et al., 2021; Shi et al., 2025).

3.3. Tectonic Marsquakes (InSight LF and BB Events)

However, an alternative interpretation is that the Viking sol 80 event did not originate from the InSight seasonal seismic region close to the Viking 2 landing site. In this scenario, the observed first-arrival phase is not a P wave, but rather a conversion phase (i.e., the so-called Sp phase) associated with a crustal discontinuity (Lazarewicz, 2023). Note that sol 80 occurred during solar conjunction (sols 64–110), when Event-mode recording was manually scheduled between local time 3:00 and 4:00. A P wave arriving within this window would have been recorded regardless of the trigger threshold. Yet, the S-wave arrival at local time ~03:00:35 implies that a P-S differential travel time exceeding 35 s (corresponding to an epicentral distance of 5°) would place the P wave outside the scheduled recording window. The absence of a recorded P arrival could therefore result from its failure to trigger Event mode when occurring outside the scheduled recording window, potentially because of either a high trigger threshold or an unfavorable radiation pattern near a nodal plane. As such, the sol 80 event would originate at a much greater epicentral distance than the originally inferred ~110 km.

In such case, it is plausible that the P-wave amplitude was too weak to trigger the Event mode, for example, if under a high trigger threshold or if near a nodal plane of the radiation pattern. Alternatively, the missing P wave could also indicate that the sol 80 event resembles InSight's single-phase events, which exhibit clear P-wave onsets but weak or absent S-wave energy (Khan et al., 2021), such as the S0183a event near the Cerberus Fossae region (Figure 1a).

Figure 3. Seismicity and waveform comparison between Earth, Moon, and Mars. (a) Cumulative seismicity as a function of moment magnitude for InSight tectonic marsquakes (low-frequency and broadband events, excluding confirmed meteorite impacts), Apollo shallow moonquakes (Oberst, 1987), and terrestrial earthquakes. The red shaded region indicates the estimated magnitude range of the Viking 2 sol 80 event's near-vertical X component for epicentral distances of 39°–85°, assuming a *b*-value of 1. Magnitude estimates are obtained by amplitude scaling with the InSight S1102a broadband event (Text S9 in Supporting Information S1). Open circles indicate the minimum detectable magnitude (amplitude ≥1 digital unit [DU]), filled circles indicate the Viking ambient noise level (amplitude = 10 digital units), and filled triangles mark the observed sol 80 amplitude (~30 digital units). Dots along the Mars and Moon curves indicate magnitude intervals containing detected events. Terrestrial global seismicity is from the GlobalCMT catalog (Ekström et al., 2012), whereas intraplate seismicity represents stable continental interiors (Okal & Sweet, 2007). (b–e) Viking Event-mode records of a California earthquake (b), a California explosive source (c), the Viking 2 sol 80 event (d), and a simulated Viking recording of the InSight high-spectral-frequency seasonal marsquake S1045a, converted to an epicentral distance of 2° relative to Viking 2 (e). The California earthquake and explosion waveforms are reproduced from Anderson et al. (1977). Anderson et al. (1977) reported three small onsets before the main peak of the sol 80 event; however, the earliest onset is likely an artifact, as only the latter two onsets are present in Lazarewicz (2023) and this study. The horizontal dashed line in (e) denotes the estimated Viking self-noise floor (Text S8 in Supporting Information S1).

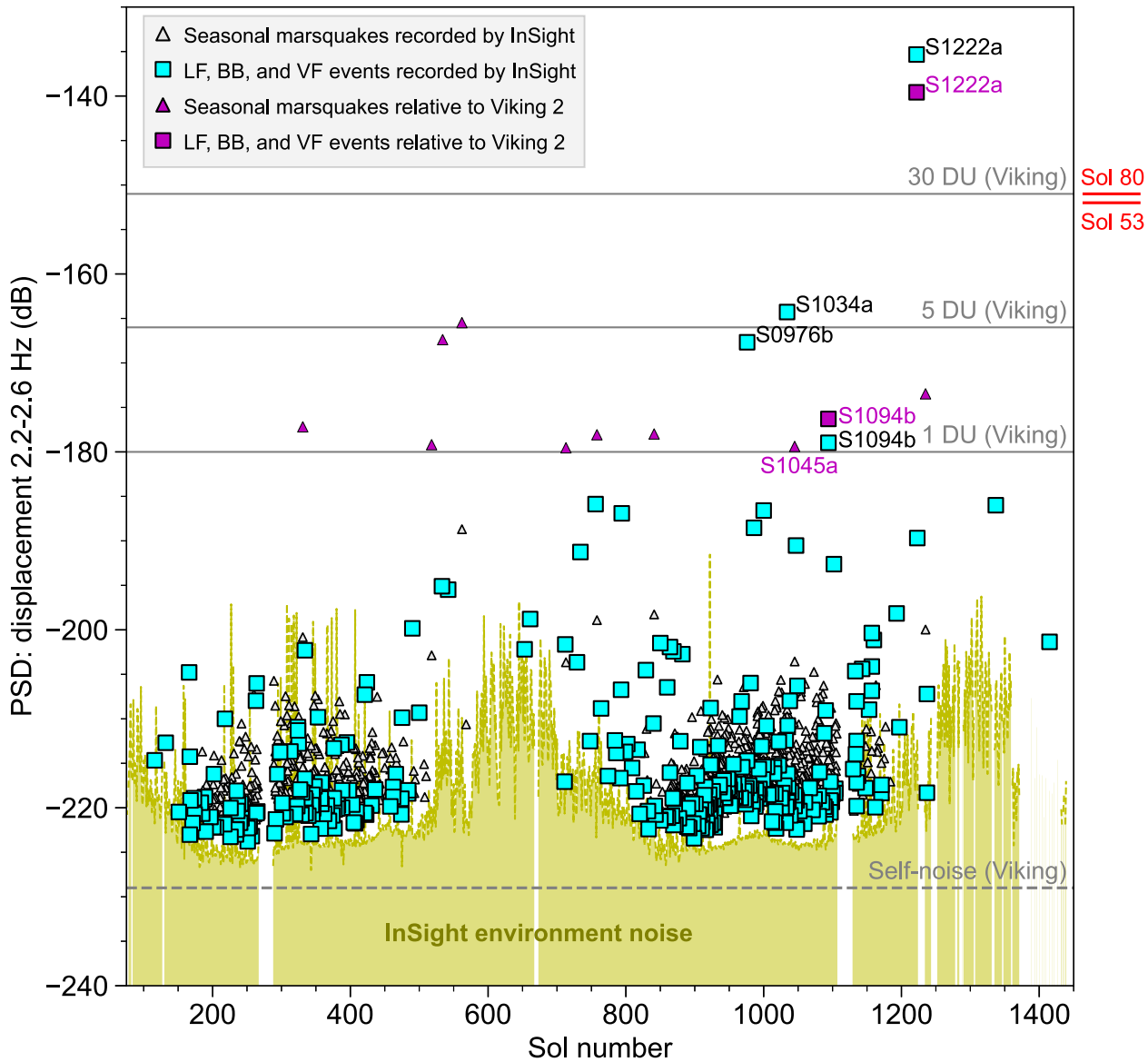


Figure 4. Reconstructing Viking 2 detectability using modern InSight marsquake observations. The *y*-axis shows the mean one-sided power spectral density (PSD) of displacement (dB), averaged over the 2.2–2.6 Hz frequency band. Yellow shadow region represents InSight's 90 percentile of noise. Gray triangles represent InSight seasonal marsquakes (high-spectral-frequency and 2.4-Hz events; Dahmen et al., 2026). Cyan squares represent InSight tectonic marsquakes (low-spectral-frequency and broadband events) and very-high-frequency (VF) events (InSight Marsquake Service, 2023). Magenta symbols represent the expected PSDs of InSight events at the Viking 2 landing site with amplitudes ≥1 digital unit (DU) after correcting the source-receiver distance from relative to InSight to relative to Viking 2 (~44° to ~2° for seasonal marsquakes). Red lines represent the DU levels of recorded Viking candidate events. Horizontal gray lines indicate the estimated Viking detection levels for various DUs and the Viking self-noise floor (Texts S7–S8 in Supporting Information S1).

If this interpretation is correct, the sol 80 event could instead belong to the distant tectonic marsquake population later observed by InSight. A key observation supporting this interpretation is the strong relationship between epicentral distance and spectral frequency among InSight tectonic marsquakes. Events originating from Cerberus Fossae at epicentral distances within ~35° typically exhibit dominant frequencies below ~1 Hz, whereas tectonic events detected at greater distances, extending to ~73°, commonly preserve high-frequency energy above ~3 Hz (Figure S28 in Supporting Information S1). This pattern likely reflects weak seismic attenuation in the Martian deep interior (Li et al., 2025), which allows high-spectral-frequency seismic energy to survive long propagation paths. Notably, the sol 80 event also exhibits unusually high spectral energy reaching 3–7 Hz, consistent with the spectral characteristics of distant tectonic marsquakes. Because the InSight and Viking 2 landing sites are located at different regions on Mars, the well-located BB marsquakes with high-spectral-frequency energy detected by

InSight at epicentral distances of ∼28°–73° relative to InSight correspond to distances of ∼39°–85° relative to Viking 2.

Under this interpretation, the originally inferred local distance for the sol 80 event (∼110 km) and its corresponding Richter magnitude (M_L 2.8) estimate would both be substantially underestimated. We therefore estimate its magnitude using amplitude scaling with the simulated Viking 2 waveform of the InSight S1102a event, a distant marsquake (73.31°, M_W 3.2) recorded by InSight with similar high-spectral-frequency energy. Specifically, we compare the simulated *X*-component (near-vertical) Viking amplitudes of S1102a with the sol 80 amplitude and apply the corresponding distance correction for an epicentral distance range of 39°–85°, yielding an estimated moment magnitude of M_W 4.6–4.8 (Figure 3a). The *Z*-component records are systematically the largest, due to the rocking motion of the Viking 2 lander as its one footpad rested on a rock (Anderson et al., 1977). Scaling the magnitude using the *Z* component yields a moderately larger estimate (M_W 4.9–5.1). The detection of sol 80 (M_W 4.6–4.8 or M_W 4.9–5.1) implies that an event larger than the largest InSight marsquake (S1222a, M_W 4.6 ± 0.2) could have occurred during the Viking 2 mission. However, the sparse detections and limited observation durations preclude a definitive comparison of seismicity rates between the two missions.

A potential question arises from the Gutenberg-Richter law, which states that the number of smaller-magnitude earthquakes increases exponentially relative to larger-magnitude events. Per this law, Viking 2 should have detected hundreds of smaller marsquakes during its 560-sol's monitoring period, which sharply contrasts with the only 1–2 events observed (Figure 3). This apparent discrepancy arises from Viking 2's lower seismic sensitivity: smaller marsquakes have amplitudes either below the instrument's digital resolution threshold (i.e., 1 DU) or close to the background noise (e.g., 10 DUs; Goins & Lazarewicz, 1979). Based on the Gutenberg-Richter law (e.g., with the maximum magnitude ranging from M_W 4.6 to 4.8), only 2 to 9 marsquakes with DUs of 1–10 are expected throughout the entire Viking 2 mission (Figure 3). Under Poisson statistics, the probability of detecting ≤3 events with 10-DU amplitude is 85.7%, and the probability of detecting ≤9 events with 1-DU amplitude is 58.7%. This already limited number of potential detections may have been missed entirely, as available Event-mode data cover only roughly one-third of Viking 2's operational lifetime (Figure 1b).

Finally, site effect also plays an important role in shaping the recorded amplitudes. The InSight landing site is known to exhibit seismic amplification at certainty frequencies (e.g., 8.0 Hz) because of shallow low-velocity sediments (Carrasco et al., 2022; Xiao et al., 2023). Similar amplification can increase signal amplitudes and enhance detectability by a low-sensitivity instrument such as Viking. In the absence of comparable site-characterization constraints for Viking 2, our magnitude estimates for the sol 80 event, derived assuming a bedrock setting, should therefore be regarded as upper bounds.

3.4. Meteorite Impacts (InSight VF Events)

Another possible interpretation for the sol 80 event is a meteorite impact, which also produces high-spectral-frequency signals (Garcia et al., 2022; Horleston et al., 2022; Posiolova et al., 2022). In addition to the LF, BB, HF, and 2.4-Hz marsquake families identified in the InSight mission, the InSight catalog also contains a smaller population of VF events. Recent studies have proposed that these VF events may predominantly originate from meteorite impacts rather than tectonic processes (Daubar et al., 2024; Zenhäusern et al., 2024). In particular, Zenhäusern et al. (2024) interpreted the VF event population as impact-generated and derived empirical crater-size versus distance scaling relationships from these events.

With this relationship, the high-spectral-frequency characteristics of the sol 80 signal are broadly consistent with a possible impact origin with a hundred-meter-scale diameter (Figure S29 in Supporting Information S1). If correct, the Viking event would provide constraints primarily on Martian impact rates rather than internal seismic activity, though a systematic search of Viking-era orbital imagery for such a crater lies beyond this study's scope.

Impacts may potentially generate infrasound signals in addition to seismic signals (Garcia et al., 2021, 2022; Xu et al., 2022). However, the simulated amplitude for InSight S0986c's infrasound signal, at a distance of 1.44°, falls well below the 1-DU detection threshold (Figure S19 in Supporting Information S1). Furthermore, the 7 Hz spectral peak of the sol 80 event is higher than the <3 Hz signals typical of impact-induced infrasound in InSight observations. The low-sampling-rate meteorological measurements of Viking are also unlikely to have recorded such a signal, especially given the absence of synchronous meteorological data during the sol 80 event.

3.5. Short-Lived Seismic Signal of Sol 80

The sol 80 event displays a remarkably short duration, decaying by a factor of $1/e^2$ (13.5%) from its maximum X-component amplitude (~30 DUs) in approximately 23 s (Figure 3d). This decay rate is 20 times faster than marsquakes recorded by InSight (Figure 3e). Although the Viking processing workflow can shorten signal duration to some extent (Figures S16–S23 in Supporting Information S1), it cannot explain the order-of-magnitude difference in durations for Viking and InSight events. Instead, the signal duration is more comparable to shallow tectonic earthquakes or explosive seismic sources on Earth (Figures 3b and 3c; Anderson et al., 1977; Horwitz et al., 1998).

This observation contrasts with the commonly held view derived from InSight that Martian seismic signals generally exhibit much longer coda and slower decay than terrestrial earthquakes. Instead, it implies heterogeneity in the attenuation structure of Mars, and suggests that terrestrial-analogous attenuation may exist in Utopia Planitia, potentially associated with spatial variations in subsurface volatile or water content (Caspari et al., 2014; Subramaniyan et al., 2015) in this heavily impact-altered region. Although such an interpretation remains speculative, it suggests that attenuation properties inferred from the InSight landing site may not necessarily represent the global or northern-hemispheric Martian crust.

4. Conclusions

Fifty years after Viking 2 returned the first seismic observations from Mars, we reconstruct its unique seismic workflow and reassess the most credible candidate event, the sol 80 signal, using knowledge gained from the InSight mission. Our analysis reveals that the apparent contrast between Viking and InSight seismic observations primarily reflects differences in instrument response, sampling strategy, and the distinct spectral characteristics of Martian seismic events, rather than a change in Martian activity alone.

The Viking 2 sol 80 event provides new constraints on Mars' seismic history. Its independently inferred location overlaps the broad northern-plains region associated with InSight seasonal marsquakes, suggesting a possible long-lived active region outside Cerberus Fossae. However, alternative origins, including a distant tectonic marsquake analogous to InSight BB events or a meteorite impact analogous to VF events, cannot be excluded.

More broadly, this study transforms the Viking seismic record from a limited early experiment into a valuable complement to InSight, extending Martian seismology across both space and time. By connecting observations separated by five decades, this work demonstrates the unique value of integrating legacy and modern planetary data sets to build longer observational records and reveal the spatial and temporal variability of planetary processes. As planetary seismology enters a new era, such integration will be essential for extracting the full scientific potential of both historical and future missions.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The Viking 2 seismic datasets are available on NASA's Planetary Data System (Lorenz et al., 2017), JAXA (https://darts.isas.jaxa.jp/en/missions/viking), and the University of Texas Institute for Geophysics dataset (https://darts.isas.jaxa.jp/en/datasets/darts:viking-utig-data). Individual meteorological data of Viking missions are available on NASA's Planetary Data System (https://pds-atmospheres.nmsu.edu/data_and_services/atmospheres_data/catalog.htm#Mars). The original InSight seismic data are available from NASA's Planetary Data System (InSight Mars SEIS Data Service, 2019a), IPGP's MSDC (InSight Mars SEIS Data Service, 2019b), and the IRIS DMC (https://www.iris.edu/hq/sis/insight). This study adopts the processed InSight seismic data (S. Wang et al., 2025) with glitch and tick noise removed. The Mars seismic catalogue is available from InSight Marsquake Service (2023).

All technical details are well described. Fortran codes in the Viking 2 seismometer's documentation are public in literature (Lazarewicz, 2023) and on Zenodo (Lazarewicz, 2022; https://doi.org/10.5281/zenodo.7272567).

Developed Python-based codes in this study are public on Zenodo (Xiao & Li, 2026; https://doi.org/10.5281/zenodo.22140586).

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